YEAR 10 SCIENCE EXTENSION
NATURAL AND PROCESSED MATERIALS
ASSIGNMENT 2: CHEMICAL CALCULATIONS

NAME: SOLUTIONS **Total:** —/34

1. Calculate the formula mass for the following substances.

NAME	FORMULA	FORMULA MASS ($gmol^{-1}$)
sodium phosphate	Na_3PO_4	163.94
iron (III) nitrate	$Fe(NO_3)_3$	241.88
calcium hydrogencarbonate	$Ca(HCO_3)_2$	162.12
barium sulphide	BaS	169.36
zinc acetate	$Zn(CH_3COO)_2$	183.47

(10)

2. Convert the following to moles or mass as necessary.

(a) 36.2 g of copper (II) nitrate.

$$n(Cu(NO_3)_2) = \frac{m}{M}$$
$$= \frac{36.2}{187.57} \quad (1)$$
$$= \underline{0.193\ mol} \quad (1)$$

(b) 2.34 moles of aluminium hydroxide.

$$m(Al(OH)_3) = n \times M$$
$$= (2.34)(78.004) \quad (1)$$
$$= \underline{182.5g} \quad (1)$$

(b) 1.34 kg of iron (III) oxide.

$$n(Fe_2O_3) = \frac{m}{M}$$
$$= \frac{1.34 \times 10^3}{159.70} \quad (1)$$
$$= \underline{8.39\ mol} \quad (1)$$

(d) 1.15×10^3 moles of sulphur trioxide.

$$m(SO_3) = n \times M$$
$$= (1.15 \times 10^3)(80.06) \quad (1)$$
$$= \underline{9.21 \times 10^4 g} \quad (1)$$

(8)

3. 27.2 g of aluminium carbonate is reacted with excess sulphuric acid. The equation for the reaction is as follows.

$$Al_2(CO_3)_{3(s)} + 3\ H_2SO_{4(aq)} \rightarrow Al_2(SO_4)_{3(aq)} + 3\ H_2O_{(l)} + 3\ CO_{2(g)}$$

✓ ? ?

(a) Calculate the mass of sulphuric acid used in the reaction?

$$n(Al_2(CO_3)_3) = \frac{m}{M}$$

$$= \frac{27.2}{233.99} \quad (1)$$

$$= 0.1162\ \text{mol} \quad (\tfrac{1}{2})$$

$$n(H_2SO_4) = \frac{3}{1} \times n(Al_2(CO_3)_3) \quad (1)$$

$$= 3(0.1162)$$

$$= 0.3486\ \text{mol} \quad (\tfrac{1}{2})$$

$$m(H_2SO_4) = n \times M$$

$$= (0.3486)(98.076) \quad (1)$$

$$= \underline{34.19\ g} \quad (1)$$

(5)

(b) What mass of carbon dioxide would be produced?

$$n(CO_2) = \frac{3}{1} \times n(Al_2(CO_3)_3) \quad (1)$$

$$= 3(0.1162)$$

$$= 0.3486\ \text{mol} \quad (\tfrac{1}{2})$$

$$m(CO_2) = n \times M$$

$$= (0.3486)(44.01) \quad (1)$$

$$= \underline{15.34\ g} \quad (\tfrac{1}{2})$$

(3)

4. Ethanol burns in air according to the following equation.

$$C_2H_5OH_{(l)} + 3\ O_{2(g)} \rightarrow 2\ CO_{2(g)} + 3\ H_2O_{(g)}$$

✓ ? ?

If 3.85×10^2 g of ethanol burns, calculate:

(a) the mass of O_2 required.

$$n(C_2H_5OH) = \frac{m}{M}$$

$$= \frac{3.85\times10^2}{46.068} \quad (1)$$

$$= 8.357\,\text{mol} \quad (\tfrac{1}{2})$$

$$n(O_2) = \frac{3}{1} \times n(C_2H_5OH) \quad (1)$$

$$= 3(8.357)$$

$$= 25.071\,\text{mol} \quad (\tfrac{1}{2})$$

$$m(O_2) = n \times M$$

$$= (25.071)(32.00) \quad (1)$$

$$= \underline{8.02\times10^2\,g} \quad (1)$$

(5)

(b) the mass of CO_2 produced.

$$n(CO_2) = \frac{2}{1} \times n(C_2H_5OH) \quad (1)$$

$$= 2(8.357)$$

$$= 16.714\,\text{mol} \quad (\tfrac{1}{2})$$

$$m(CO_2) = n \times M$$

$$= (16.714)(44.01) \quad (1)$$

$$= \underline{7.36\times10^2\,g} \quad (\tfrac{1}{2})$$

(3)